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QNMF: A quantum neural network based multimodal fusion system for intelligent diagnosis

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ABSTRACT

The Internet of Medical Things (IoMT) has emerged as a significant research area in the medical field, enabling the transmission of various types of data to the cloud for analysis and diagnosis. Fusing data from multiple modalities can enhance accuracy but requires substantial computing power. Theoretically, quantum computers can rapidly process large volumes of high-dimensional medical data. Despite accelerated developments in quantum computing, research on quantum machine learning (QML) for multimodal data processing remains limited. Considering these factors, this paper presents a quantum neural network-based multimodal fusion system for intelligent diagnosis (QNMF) that can process multimodal medical data transmitted by IoMT devices, fuse data from different modalities, and improve the performance of intelligent diagnosis. This system employs a quantum convolutional neural network (QCNN) to efficiently extract features from medical images. These QCNN-based features are then fused with other modality features (such as blood test results or breast cell slices), and used to train an effective variational quantum classifier (VQC) for intelligent diagnosis. The experimental results demonstrate that a QCNN can effectively extract image data features. Furthermore, QNMF achieved an accuracy of 97.07% and 97.61% on breast cancer diagnosis and Covid-19 diagnosis experiments, respectively. In addition, the QNMF exhibits strong quantum noise robustness.

1. Introduction

With technological advancements, the development of smart healthcare systems has become a crucial aspect of contemporary society. This trend not only accelerates the digitization and intelligent transformation of the medical industry but also imposes greater demands on future medical services. To deliver more efficient and accurate care, these services need to incorporate advanced technologies, such as the Internet of Things (IoT) [1], artificial intelligence (AI) [2], and quantum computing [3]. The development of smart healthcare systems requires numerous Internet of Medical Things (IoMT) devices to enable real-time patient monitoring and vital-sign collection, thereby enhancing the efficiency of medical professionals. Through integration with AI, the medical industry can elevate overall service quality and treatment standards, and alleviate strained doctor-patient relationships. The data collected by IoMT devices are used to train a neural network for intelligent diagnosis that assists physicians in clinical decision-making, helps doctors make informed diagnoses and avoids experiential pitfalls, boosts the accuracy of clinical decisions and significantly reduces misdiagnoses.

Nonetheless, data gathered by IoMT devices often appear in multiple forms or originate from various sources, and are typically referred to as multimodal data. Multimodal data processing is a critical challenge in medical data analysis. Given the robust representation and fusion capabilities of deep learning (DL) methods, DL technology has emerged as a crucial approach to address heterogeneous data processing in increasingly intricate multimodal learning. DL-based multimodal fusion systems utilize DL models to extract features from diverse data types, such as text, images, speech, and videos, to perform fusion and further efficient detection/recognition tasks [4]. In contrast to unimodal learning, which relies solely on a single-mode data type for training, multimodal learning has significant advantages in terms of modeling ambiguity and uncertainty from the input data.

However, conventional DL fusion methods encounter several issues when handling multimodal data: first, the data dimensions become

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considerably large; and second, an increasing amount of multimodal data must be processed, substantially increasing the computational power requirements. Quantum computing and machine learning can theoretically effectively address these problems. Quantum computers leverage quantum parallelism to rapidly process vast quantities of medical data, and the properties of quantum superposition states enable more efficient high-dimensional data processing using fewer quantum computing resources, effectively mitigating the severe computational power shortage currently faced by AI. However, research on quantum machine learning (QML) for multimodal medical data fusion is significantly lacking, necessitating further development. Therefore, this study proposes a system that employs a quantum convolutional neural network (QCNN) to effectively fuse multimodal medical data and apply it to train a variational quantum classifier (VQC) for intelligent diagnosis, called QNMF.

The main contributions of this paper are as follows:

- A quantum neural network-based multimodal fusion system for intelligent diagnosis is proposed.
- A quantum convolutional neural network is proposed to extract image features effectively. It was optimized using a classical backpropagation algorithm and exhibited robustness against quantum noise.
- This study presents a comprehensive mathematical proof of the adaptability of the classical backpropagation algorithm to the proposed QCNN.

The remainder of this paper is organized as follows: Section 2 presents the relevant background knowledge. Section 3 describes the QNMF system. Section 4 describes the experiments conducted using QNMF to process medical data on the Pennylane [5] platform and analyzes the impact of quantum noise on the model. Finally, Section 5 concludes the paper and discusses future research directions.

2. Related work

This section introduces the research progress related to this study, including the development of AI-driven smart healthcare technology and achievements of quantum neural network algorithms.

2.1. AI-driven smart healthcare system

During the initial phase of rapid advancements in AI, numerous efforts were made to apply AI technology in the medical domain. Interest in smart healthcare has continued to rise in recent years. Numerous studies have been conducted on medical AI. In 2013, the renowned IBM Watson [6] could recognize natural language within the domain of cancer treatment and screen 1.5 million patient records in just a few seconds, offering doctors evidence-based treatment options. In 2017, DeepMind [7] demonstrated the application of blockchain technology to track personal health data and address patient privacy concerns.

Nonetheless, a significant portion of data in smart healthcare is multimodal. Because information across various modalities is often redundant or complementary, multimodal learning has emerged as a crucial subfield of AI. Alshehri et al. [8] provided a comprehensive overview of recent advancements in smart healthcare and research challenges for future development. The authors emphasize that data fusion is a crucial direction for the future evolution of smart healthcare. Muhammad et al. [9] summarized the work on data fusion in smart healthcare over the past decades and provided valuable insights and implications for its future development. Chen et al. [10] emphasized the importance of enhancing the confidence and motivation of professionals/medical experts to integrate AI-driven information fusion systems into daily clinical practice. They also highlighted the necessity of creating a real-time decision support system to evaluate the adaptability in end-to-end practical diagnostic scenarios.

In 2014, Kiros et al. [11] introduced a feature-based multimodal DL approach for tissue and organ segmentation by incorporating features into classifiers or energy-based segmentation algorithms. This filled a research gap because most previous studies relied on application-specific manual feature extraction. In 2016, Yoo et al. [12] employed the Euclidean distance transformation method to augment relatively sparse image features and fused them with several clinical diagnostic features, attaining superior accuracy in predicting multiple sclerosis compared with other techniques. In 2018, Cheng et al. [13] proposed an image fusion method using DL to effectively fuse patient CT and MRI images, thereby enhancing the accuracy of intelligent diagnostic models. In the same year, Shahdoosti et al. [14] utilized DL to determine the fusion coefficients of medical images, and demonstrated that the coefficients obtained through ML methods can improve the reliability and feasibility of fused images. Later, Kawahara et al. [15] trained a convolutional neural network (CNN) by using multimodal data for skin lesion diagnosis. Because the network was trained with multiple multitask loss functions considering different input modality combinations, the model demonstrated robustness when dealing with missing data. In 2018, Reda et al. [16] developed ML models for each data modality separately and then built another classifier using the predicted probabilities of single-modality models as input, achieving 94.4% diagnostic accuracy on real medical data. In 2019, Nie et al. [17] used feature engineering techniques to calculate multiple metric maps from multimodal MRI images of gliomas, extracted shallow image features, and utilized a CNN to extract deep MRI image features. Eventually, they fused other relevant statistical information and applied support vector machines (SVM) for glioma diagnosis, achieving higher accuracy than others. In 2020, Tan et al. [18] proposed a multimodal medical image fusion algorithm based on energy attributes and a pulse-coupled neural network fusion strategy, that could be used for several medical image diagnostic problems. Ali et al. [19] proposed a smart healthcare system for predicting heart disease using an integrated feature fusion approach. It combined sensor data and features extracted from electronic medical records to generate valuable medical data. Subsequently, they selected essential features from the fused features, reducing the computational burden and enhancing system performance. Furthermore, they calculated specific feature weights for each class, and the model achieved 98.5% accuracy. In 2023, Islam et al. [20] introduced a multilevel feature fusion technique for multimodal human activity recognition, using a multi-head CNN and convolutional block attention modules to process visual data and long short term memory (LSTM) to process time-sensitive multisource sensor information. The performance of the proposed fusion neural network in real smart healthcare application scenarios surpassed that of the other methods.

2.2. Quantum Machine Learning (QML)

Quantum neural networks (QNN) use qubits and gates for computation and learning in quantum-computing environments. Unlike the classical neural network model, the QNN leverages the superposition and entanglement properties of quantum states for information processing, yielding enhanced computational efficiency and more robust information processing capabilities.

The earliest research on QML can be traced back to the QNN in 1995 when Kak et al. [21] introduced the concept of quantum neural computation. Subsequently, numerous researchers have proposed QNN models. Ventura et al. [22] suggested using quantum superposition states to represent networks, highlighting that quantum computing employs microscopic quantum effects to perform computational tasks and produces results several times faster than its classical counterparts in some cases. In 2000, Matsui et al. [23] used quantum dynamics to describe neural states and explored quantum bit neural networks corresponding to quantum computing. In 2006, Zhou et al. [24] proposed a quantum perceptron that leveraged the quantum phase and possessed computational capabilities unattainable by

traditional perceptrons. These networks aim to combine quantum mechanics with different types of data; however, the primary challenge is the inability to implement the quantum evolution of neural networks. Subsequently, researchers attempted to use quantum-computing parallelism to accelerate the training speed of classical ML. In 2013, Lloyd et al. [25] applied the superposition property to the representation of classical vectors and proposed a quantum K-means algorithm that can achieve more efficient clustering than classical K-means. In 2014, Wiebe et al. [26] used quantum state probability amplitudes to represent classical vectors and suggested a quantum nearest neighbor algorithm by comparing the distance between two quantum states. In the same year, Rebentrost et al. [27] used the density matrices of quantum systems to represent kernel matrices in support vector machines. Subsequently, Wiebe et al. [28] introduced the concept of quantum deep learning, and attempted to combine quantum computing with classical deep learning to accelerate the training of deep neural networks. In 2022, Tiwari et al. [29] proposed a quantum kernel method by mapping the input dataset to nonlinear coherent states and employed a kernel function with SVM to perform classification tasks. The experimental results demonstrate the superiority of the proposed method over the baselines, particularly under conditions of high noise.

However, most of the aforementioned studies have theoretical characteristics, and several algorithms cannot be implemented at the current stage. In the noisy intermediate scale quantum (NISQ) era [30], variational quantum algorithms [31] have become an important research direction, with QNN based on quantum circuits receiving widespread attention. This type of QNN consists of several quantum gates and optimizes the parameters of these quantum gates to achieve specific tasks. Hybrid quantum algorithms have already demonstrated some advantages over classical algorithms. In 2018, Farhi et al. [32] proposed a quantum neural network by optimizing the parameters of quantum circuits to perform binary classification tasks. Schuld et al. [33] refined this neural network structure and proposed a circuit-centric quantum classifier, making the construction of quantum neural networks easier. In 2019, Cong et al. [34] proposed a QCNN that can process classical image data while producing results similar to those of classical convolutions, thereby considerably expanding the range of QNN applications. Pesah et al. [35] proved that by constructing reasonable quantum circuits, QNN would not suffer from the barren plateaus that trouble classical neural networks, thereby demonstrating the potential advantages of QNN. In 2021, Du et al. [36] conducted classification tasks using a QNN with depolarizing noise, demonstrating that the QNN can attain robustness against attackers, which increase as the noise level intensifies. Furthermore, the authors demonstrated the model robustness is highly sensitive to the details of the classification model in classical cases. However, in quantum cases, the robustness of the classification model is insensitive to model details, providing potential quantum advantages for QNN. In 2022, Qu et al. [37] proposed a spatio-temporal quantum graph CNN to address the traffic congestion prediction (TCP) problem. The results show that the quantum algorithm can effectively solve the TCP problem and exhibits resistance to interference. In addition, in 2023, Mensa et al. [38] combined classical support vector classifier algorithms with ligand-based virtual screening (LB-VS) quantum kernel estimation in real databases to achieve potential quantum advantages. The authors heuristically demonstrated that in some cases, quantum algorithms can provide advantages over most classical algorithms. Quantum algorithms play a significant role in smart healthcare systems. In 2022, Laxminarayana et al. [39] proposed a hybrid QNN model for intrusion detection in healthcare that successfully outperformed classical models, such as CNN and LSTM. In 2023, Karthick et al. [40] designed an intelligent machine-learning forecasting framework (IMLFF) for diagnosing of chronic obstructive pulmonary disease (COPD). This framework employs a fuzzy-based QNN classifier for COPD diagnosis. The experimental results demonstrated that the IMLFF outperformed the existing methods in various statistical parameters and performance metrics. In the same year, Nijaguna et al. [41] used the Quantum Fruit Fly Algorithm (QFFA) technique for feature selection to improve classification effectiveness in medical diagnosis. This study applied a new normalization technique to normalize images, reduce the differences in pixel values, and enhance the images. The QFFA model outperforms most existing deep 1D-CNN and GA-Decision Tree models in various performance metrics.

Despite the rapid development of QML, limited research has been conducted on data fusion in the medical domain. Moreover, several related studies did not employ NISQ-era algorithms or consider the impact of noise on these algorithms. Therefore, it is crucial to design a QNN-based multimodal fusion system using medical data that can be implemented in current application environment.

3. Proposed QNMF system

The QNMF system comprises several stages: data collection and preprocessing, feature extraction, feature fusion, and training of a VQC for intelligent diagnosis (Fig. 1). First, during the stage of data collection and preprocessing, pre-trained QCNNs were utilized to extract features from specific modality data gathered by the IoMT (Section 3.1). Second, an early fusion approach [42] was employed in the feature fusion stage to fuse the images and EHR data and generate the features used to train the VQC (Section 3.2). Third, a VQC for intelligent diagnosis was trained using the fused features (Section 3.3). Each stage is described in detail.

3.1. Data collection and preprocessing

Multimodal patient data were collected using corresponding IoMT devices. Furthermore, the collected data were screened, labeled, and uploaded to the cloud for data preparation for subsequent processes.

We employed a trainable quantum circuit to extract features from the specific modality data. By adding a pooling layer or a fully connected layer, it can achieve functionality similar to that of a classical CNN, referred to as QCNN. Particularly, in QNMF, we pre-train a QCNN for image feature extraction, which included the following processes: data encoding, quantum evolution, data decoding, and a pre-trained model for feature extraction.

3.1.1. Data encoding

The primary goal of data encoding is to encode a classical patch ($\boldsymbol{x}$) into a quantum state using an encoding circuit. This process involves the evolution of the quantum pure state $|0\rangle^n$ into $|\boldsymbol{x}\rangle$ through $U_{encoding}(\boldsymbol{x})$.

In general, a few data encoding methods are available, including *direct encoding* that directly encodes classical 0 and 1 bits into qubits $|0\rangle$ and $|1\rangle$, *amplitude encoding* [29,43,44] that normalizes classical data and stores it in the amplitude of quantum states, and *angle encoding*. The core idea of angle encoding is to represent the input data as the angles of the quantum rotation gates. In particular, angle encoding applies a unitary transformation to a pure quantum state to obtain an input quantum state. Its mathematical representation is given by Eq. (1).

$$U_{encoding}(\boldsymbol{x})|0\rangle^n = |\boldsymbol{x}\rangle. \tag{1}$$

Here, $\boldsymbol{x} = \left(x_1, x_2, \ldots x_n\right)^T$ is the input for quantum feature extraction and n is the patch size, which is usually chosen as 2×2 or 3×3. $U_{encoding}(\boldsymbol{x})$ is a unitary transformation that encodes data in a quantum state. The mathematical expression for $U_{encoding}(\boldsymbol{x})$ is Eq. (2).

$$U_{encoding}(\boldsymbol{x}) = \bigotimes_{i=1}^{n} R_y(\pi x_i). \tag{2}$$

Here, R_y refers to a quantum rotation gate.

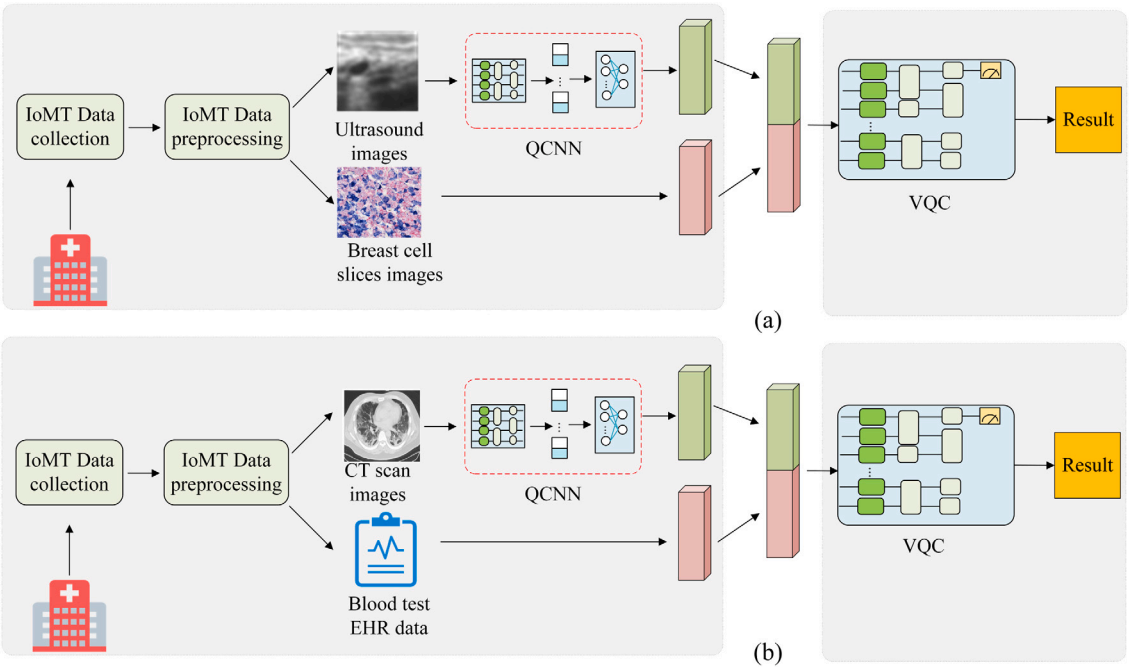


Fig. 1. Flowchart of QNMF framework for the different tasks. (a) QNMF for breast cancer diagnosis, and (b) QNMF for Covid-19 diagnosis. (QCNN: Quantum Convolutional Neural Network, VQC: Variational Quantum Classifier, EHR: Electronic Health Records).

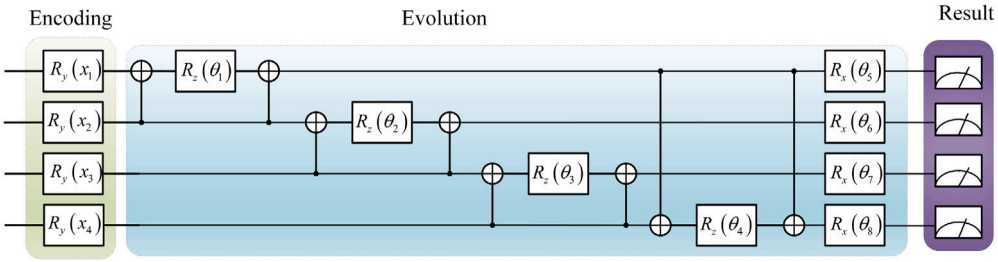


Fig. 2. The ansatz for QCNN.

3.1.2. Quantum evolution

After encoding a classical patch into a quantum state $|\boldsymbol{x}\rangle$ through the encoding circuit, another unitary transformation U_{evo} is applied to evolve it into another quantum state $|\boldsymbol{x};\boldsymbol{\theta}\rangle$, for entanglement of the information and extracting image features later. The mathematical representation is given by Eq. (3).

$$U_{evo}(\boldsymbol{\theta})|\boldsymbol{x}\rangle = |\boldsymbol{x};\boldsymbol{\theta}\rangle. \tag{3}$$

Here, $U_{evo}(\boldsymbol{\theta})$ is referred to as the ansatz or quantum convolutional layer. Choosing a powerful ansatz is key to improving the accuracy of the quantum algorithm. In theory, by increasing the number of layers and quantum gates in the ansatz, the expressive power of the ansatz can be enhanced, thus improving the performance of the QCNN. However, as the number of layers increases, the factors such as quantum noise may cause gradient vanishing problems and longer training time. Generally, by using the $\{H, R_X, R_Y, R_Z, CNOT, \mathrm{CZ}\}$ gates, any unitary transformations can be simulated. The ansatz for QCNN used in this paper is shown in Fig. 2, and the detailed process is shown as Algorithm 1.

3.1.3. Data decoding

After quantum evolution, $|\boldsymbol{x};\boldsymbol{\theta}\rangle$ was obtained. Data decoding involves measuring an expectation value of a specific observable $\hat{H}$ of the quantum state and denoting this transformation as U_M. The output of the quantum circuit can be expressed using Eq. (4).

$$U_M|\boldsymbol{x};\boldsymbol{\theta}\rangle = Tr\left(|\boldsymbol{x};\boldsymbol{\theta}\rangle\langle\boldsymbol{x};\boldsymbol{\theta}|\hat{H}\right). \tag{4}$$

Here, $|\boldsymbol{x};\boldsymbol{\theta}\rangle$ is the quantum state after the evolution, Tr is the trace operator, and $\hat{H}$ is an arbitrary observable unitary operator. The default measurement was $\hat{Z}$ basis measurement. The measurement results are then fed into a classical fully connected network to obtain the model output, which is used to optimize the parameters.

3.1.4. The pretrained model for feature extraction

The training process for QCNN is illustrated in Fig. 3. A classical CNN can be trained using a backpropagation algorithm; however, this training strategy cannot be directly applied to train a QCNN. This is because the output of the QCNN must be obtained by measuring the quantum states, which is quite different from the output of a classical CNN. Therefore, this study proposes a training algorithm for calculating the gradient of the quantum convolutional layer using the parameter shift rule and chain rule [45,46], and embeds it into the standard backpropagation process. This allows for the training of a QCNN as a whole and the relevant discussion is presented in the following section. We pre-trained a QCNN for separate image data feature extraction.

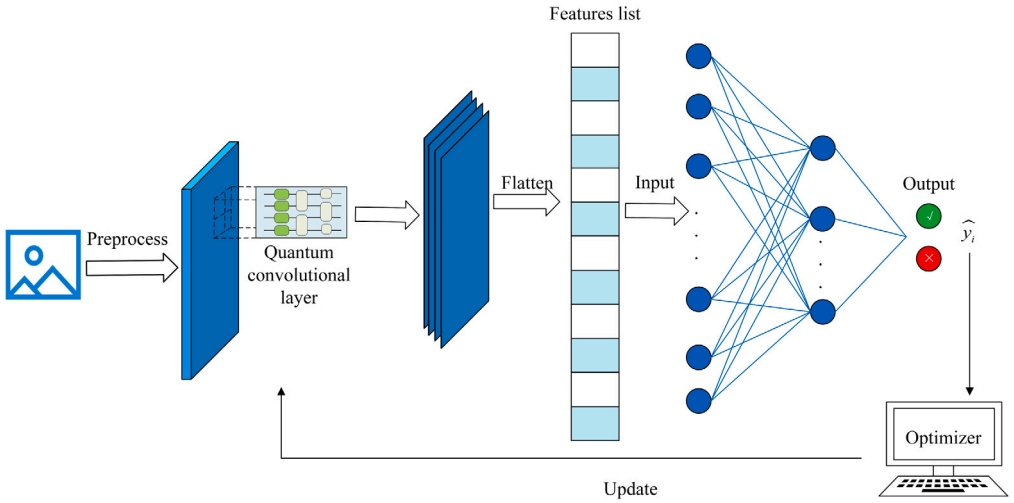


Fig. 3. The training process of the QCNN.

Algorithm 1 Construction of the ansatz for QCNN.

Input: the number of layers l, the input patch $\boldsymbol{x}$, the number of qubits n.
Output: the ansatz for QCNN.

Set $i = 1$;
for $i \leq n$ **do**
  add R_x on qubit i with parameter $\{\alpha_i = \pi x_i\}$;
  $i = i + 1$
end for
Set $i = 1$;
for $i \leq n$ **do**
  add $CNOT$ gate on qubit i with control qubit $(i + 1)\,\%n$;
  $i = i + 1$;
end for
Set $i = 1$;
for $i \leq n$ **do**
  add R_z on qubit i;
  $i = i + 1$;
end for
Repeat step 6 to 10;
Set $i = 1$;
for $i \leq n$ **do**
  add R_y on qubit i;
end for
  $i = i + 1$;
end for
Repeat step 6 to 10;
Set $i = 1$;
for $i < l$ **do**
  Repeat step 1 to 22;
end for
return circuit

The required dimensional image features are obtained by removing the end-to-end fully connected layer.

3.1.5. Backpropagation process for QCNN

As mentioned in Section 3.1.4, a significant difference is observed between the results obtained from the QCNN and classical CNN. This implies that traditional methods may not be suitable for QCNN. Therefore, adopting a backpropagation algorithm suitable for QCNN is necessary.

First, let the output of the quantum convolutional layer be $E(\boldsymbol{\theta};\boldsymbol{x})$, input quantum state $|0\rangle^n$ be $|0\rangle$, encoding process be $U(\boldsymbol{x})$, and entire evolution be $U(\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ is the parameter of the quantum convolution kernel. The output quantum state can be written as $U(\boldsymbol{\theta})U(\boldsymbol{x})|0\rangle = |\boldsymbol{x};\boldsymbol{\theta}\rangle$, and the $\hat{Z}$ basis measurement result can be written as Eq. (5).

$$E(\boldsymbol{\theta};\boldsymbol{x}) = Tr\left(|\boldsymbol{x};\boldsymbol{\theta}\rangle\langle\boldsymbol{x};\boldsymbol{\theta}|\hat{Z}\right) = \langle 0|U^{\dagger}(\boldsymbol{x})U^{\dagger}(\boldsymbol{\theta})\hat{Z}U(\boldsymbol{\theta})U(\boldsymbol{x})|0\rangle. \tag{5}$$

To calculate the subsequent process, we need to calculate $\frac{\partial o_{uv}}{\partial \theta_l}$ and $\frac{\partial o_{uv}}{\partial x_{ij}}$.

Let us first calculate $\frac{\partial o_{uv}}{\partial \theta_l}$. Based on the description in this context, o_{uv} is the measurement value after evolution and can be directly replaced with $E(\boldsymbol{\theta};\boldsymbol{x})$. Furthermore, a unitary transformation can be decomposed into a combination of smaller unitary transformations, as Eq. (6).

$$U(\boldsymbol{\theta}) = U_n(\theta_n)\,U(\theta_{n-1})\cdots U_0(\theta_0). \tag{6}$$

For $E(\boldsymbol{\theta};\boldsymbol{x})$, we have Eq. (7).

$$\begin{aligned}
\frac{\partial E(\boldsymbol{\theta};\boldsymbol{x})}{\partial \theta_l} &= \frac{\partial\left\langle 0|U^{\dagger}(\boldsymbol{x})U^{\dagger}(\boldsymbol{\theta})\hat{Z}U(\boldsymbol{\theta})U(\boldsymbol{x})|0\right\rangle}{\partial \theta_l} \\
&= \langle 0|U^{\dagger}(\boldsymbol{x})\frac{\partial U^{\dagger}(\boldsymbol{\theta})}{\partial \theta_l}\hat{Z}U(\boldsymbol{\theta})U(\boldsymbol{x})|0\rangle \\
&+ \langle 0|U^{\dagger}(\boldsymbol{x})U^{\dagger}(\boldsymbol{\theta})\hat{Z}\frac{\partial U(\theta_l)}{\partial \boldsymbol{\theta}}U(\boldsymbol{x})|0\rangle \\
&= \langle 0|U^{\dagger}(\boldsymbol{x})U_0^{\dagger}(\theta_0)\cdots\frac{\partial U_l^{\dagger}(\theta_l)}{\partial \theta_l}\cdots U_n^{\dagger}(\theta_n)\hat{Z}U(\boldsymbol{\theta})U(\boldsymbol{x})|0\rangle \\
&+ \langle 0|U^{\dagger}(\boldsymbol{x})U^{\dagger}(\boldsymbol{\theta})\hat{Z}U_0(\theta_0)\cdots\frac{\partial U_l(\theta_l)}{\partial \theta_l}\cdots U_n(\theta_n)U(\boldsymbol{x})|0\rangle
\end{aligned} \tag{7}$$

Because $U = e^{i\alpha}R_{\hat{n}}(\theta)$ and the global phase does not have a direct impact on the measurement outcomes, we ignore the global phase. Any single-qubit unitary transformation can then be regarded as a specific angle of rotation θ around a rotation axis in the Bloch sphere. For rotation gates, the following conclusions can be drawn Eq. (8).

$$\begin{aligned}
\frac{\partial R_{\hat{n}}(\theta;x)}{\partial \theta} &= \frac{1}{2}R_{\hat{n}}\left(\theta+\frac{\pi}{2};x\right) \\
\frac{\partial R_{\hat{n}}^{\dagger}(\theta;x)}{\partial \theta} &= -\frac{1}{2}R_{\hat{n}}\left(\theta-\frac{\pi}{2};x\right).
\end{aligned} \tag{8}$$

After substituting Eq. (7), we obtain Eq. (9).

$$\begin{aligned}
\frac{\partial E(\boldsymbol{\theta};\boldsymbol{x})}{\partial\theta_l} &= \frac{\partial\left\langle 0|U^{\dagger}(\boldsymbol{x})U^{\dagger}(\boldsymbol{\theta})\hat{Z}U(\boldsymbol{\theta})U(\boldsymbol{x})|0\right\rangle}{\partial\theta_l} \\
&= \langle 0|U^{\dagger}(\boldsymbol{x})U_0^{\dagger}(\theta_0)\cdots\frac{\partial U_l^{\dagger}(\theta_l)}{\partial\theta_l}\cdots U_n^{\dagger}(\theta_n)\hat{Z}U(\boldsymbol{\theta})U(\boldsymbol{x})|0\rangle \\
&+ \langle 0|U^{\dagger}(\boldsymbol{x})U^{\dagger}(\boldsymbol{\theta})\hat{Z}U_0(\theta_0)\cdots\frac{\partial U_l(\theta_l)}{\partial\theta_l}\cdots U_n(\theta_n)U(\boldsymbol{x})|0\rangle \\
&= \frac{1}{2}(-\langle 0|U^{\dagger}(\boldsymbol{x})U_0^{\dagger}(\theta_0)\cdots U_l^{\dagger}\left(\theta_l-\frac{\pi}{2}\right)\cdots U_n^{\dagger}(\theta_n)\hat{Z}U(\boldsymbol{\theta})U(\boldsymbol{x})|0\rangle \\
&+ \langle 0|U^{\dagger}(\boldsymbol{x})U^{\dagger}(\boldsymbol{\theta})\hat{Z}U_0(\theta_0)\cdots U_l\left(\theta_l+\frac{\pi}{2}\right)\cdots U_n(\theta_n)U(\boldsymbol{x})|0\rangle) \\
&= \frac{1}{2}\langle 0|U^{\dagger}(\boldsymbol{x})U^{\dagger}\left(\theta_l^{+}\right)\hat{Z}U\left(\theta_l^{+}\right)U(\boldsymbol{x})|0\rangle \\
&- \frac{1}{2}\langle 0|U^{\dagger}(\boldsymbol{x})U^{\dagger}\left(\theta_l^{-}\right)\hat{Z}U\left(\theta_l^{-}\right)U(\boldsymbol{x})|0\rangle.
\end{aligned} \tag{9}$$

After rewriting this in the form of the expectation E, we we obtain Eq. (10).

$$\frac{\partial o_{uv}}{\partial\theta_l} = \frac{\partial E(\boldsymbol{\theta};\boldsymbol{x})}{\partial\theta_l} = \frac{1}{2}\left(E\left(\theta_l^{+};\boldsymbol{x}\right)-E\left(\theta_l^{-};\boldsymbol{x}\right)\right). \tag{10}$$

Here, $\theta_l^{\pm}=\theta\pm\frac{\pi}{2}e_l$, and e_l is the l-th unit vector in parameter space.

Similarly, to calculate $\frac{\partial o_{uv}}{\partial x_{ij}}$, we write the input of a patch as a one-dimensional column vector, and the original goal is to calculate $\frac{\partial o_{uv}}{\partial x_i}$. To simplify the calculation process, we represent $U^{\dagger}(\boldsymbol{\theta})\hat{Z}U(\boldsymbol{\theta})$ as a new observable $\hat{H}$. Subsequently, it becomes $E(\boldsymbol{\theta};\boldsymbol{x})=E(\boldsymbol{x})=Tr\left(|\boldsymbol{x}\rangle\langle\boldsymbol{x}|\hat{H}\right)=\langle 0|U^{\dagger}(\boldsymbol{x})\hat{H}U(\boldsymbol{x})|0\rangle$. Using the same method, we obtain Eq. (11).

$$\frac{\partial o_{uv}}{\partial x_{ij}} = \frac{\partial E(\boldsymbol{x})}{\partial x_i} = \frac{1}{2}\left(E\left(\boldsymbol{\theta};\boldsymbol{x}_i^{+}\right)-E\left(\boldsymbol{\theta};\boldsymbol{x}_i^{-}\right)\right). \tag{11}$$

Here, $\boldsymbol{x}_i^{\pm}=\boldsymbol{x}_i\pm\frac{\pi}{2}\boldsymbol{e}_i$, and $\boldsymbol{e}_i$ is the i-th unit vector in parameter space.

Based on the above discussion, the QCNN proposed in this study can be optimized using classical backpropagation.

First, the input is represented as $X_{n\times n}$, and the output after the quantum convolution kernel is $O_{m\times m}$. Each element o_{uv} in the output can be expressed as a specific $E(\boldsymbol{x};\boldsymbol{\theta})$. Here, $\boldsymbol{x}$ denotes the corresponding patch, $\boldsymbol{\theta}$ denotes the corresponding circuit parameter, and $m=\left(\left\lfloor\frac{n-k}{s}\right\rfloor+1\right)$. The convolution kernel size is k, and the stride is s. For simplicity, let us set s to 1, and denote the loss function as L. Assuming that $\frac{\partial L}{\partial O}$ is known, both $\frac{\partial o_{uv}}{\partial\theta_l}$ and $\frac{\partial o_{uv}}{\partial x_{ij}}$ can be obtained through the shift rule. In view of classical CNN, the demonstration that QCNN can adopt the backpropagation method requires proving that both $\frac{\partial L}{\partial\theta}$ and $\frac{\partial L}{\partial X}$ are solvable. For $\frac{\partial L}{\partial\theta}$, we have Eqs. (12) and (13).

$$\frac{\partial L}{\partial\theta} = \frac{\partial L}{\partial O}\times\frac{\partial O}{\partial\theta}, \tag{12}$$

$$\frac{\partial L}{\partial O} = \begin{pmatrix} \frac{\partial L}{o_{11}} & \frac{\partial L}{o_{12}} & \cdots & \frac{\partial L}{o_{1m}} \\ \frac{\partial L}{o_{21}} & \frac{\partial L}{o_{22}} & \cdots & \frac{\partial L}{o_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial L}{o_{m1}} & \frac{\partial L}{o_{m2}} & \cdots & \frac{\partial L}{o_{mm}} \end{pmatrix}. \tag{13}$$

To facilitate processing, the results were converted into a one-dimensional vector as shown in Eq. (14).

$$\frac{\partial L}{\partial\overline{O}} = \begin{pmatrix} \frac{\partial L}{\partial\overline{o}_1} \\ \frac{\partial L}{\partial\overline{o}_2} \\ \vdots \\ \frac{\partial L}{\partial\overline{o}_{m*m}} \end{pmatrix}. \tag{14}$$

Here, $\overline{O}$ represents the converted vector, and $\overline{o}_i$ represents o_{ij}. Thereafter, let us calculate $\frac{\partial L}{\partial\theta_l}$ as Eq. (15).

$$\begin{aligned}
\frac{\partial L}{\partial\theta_l} &= \sum_{i=1}^{m}\sum_{j=1}^{m}\frac{\partial L}{\partial o_{ij}}\frac{\partial o_{ij}}{\partial\theta_l} = \sum_{i=1}^{m\times m}\frac{\partial L}{\partial\overline{o}_i}\frac{\partial\overline{o}_i}{\partial\theta_l} \\
&= \begin{pmatrix} \frac{\partial\overline{o}_1}{\partial\theta_l} & \frac{\partial\overline{o}_2}{\partial\theta_l} & \cdots & \frac{\partial\overline{o}_{m\times m}}{\partial\theta_l} \end{pmatrix}\begin{pmatrix} \frac{\partial L}{\partial\overline{o}_1} \\ \frac{\partial L}{\partial\overline{o}_2} \\ \vdots \\ \frac{\partial L}{\partial\overline{o}_{m\times m}} \end{pmatrix} = \frac{\partial\overline{O}^{T}}{\partial\theta_l}\frac{\partial L}{\partial\overline{O}}
\end{aligned}. \tag{15}$$

So that we can have Eq. (16).

$$\frac{\partial L}{\partial\boldsymbol{\theta}} = \begin{pmatrix} \frac{\partial L}{\partial\theta_1} \\ \frac{\partial L}{\partial\theta_2} \\ \vdots \\ \frac{\partial L}{\partial\theta_k} \end{pmatrix} = \frac{\partial\overline{O}^{T}}{\partial\boldsymbol{\theta}}\frac{\partial L}{\partial\overline{O}}. \tag{16}$$

Here, $\frac{\partial\overline{O}^{T}}{\partial\theta} = \begin{pmatrix} \frac{\partial\overline{o}_1}{\partial\theta_1} & \frac{\partial\overline{o}_2}{\partial\theta_1} & \cdots & \frac{\partial\overline{o}_{m\times m}}{\partial\theta_1} \\ \frac{\partial\overline{o}_1}{\partial\theta_2} & \frac{\partial\overline{o}_2}{\partial\theta_2} & \cdots & \frac{\partial\overline{o}_{m\times m}}{\partial\theta_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial\overline{o}_1}{\partial\theta_k} & \frac{\partial\overline{o}_2}{\partial\theta_k} & \cdots & \frac{\partial\overline{o}_{m\times m}}{\partial\theta_k} \end{pmatrix}$. Since $\frac{\partial o_{uv}}{\partial\theta_l}$ is known, this indicates that $\frac{\partial L}{\partial\theta}$ is solvable. Next, we calculate $\frac{\partial L}{\partial X}$, which first requires calculating $\frac{\partial O}{\partial X}$ using Eq. (17).

$$\frac{\partial O}{\partial X} = \begin{pmatrix} \frac{\partial O}{\partial x_{11}} & \frac{\partial O}{\partial x_{12}} & \cdots & \frac{\partial O}{\partial x_{1n}} \\ \frac{\partial O}{\partial x_{21}} & \frac{\partial O}{\partial x_{22}} & \cdots & \frac{\partial O}{\partial x_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial O}{\partial x_{n1}} & \frac{\partial O}{\partial x_{n2}} & \cdots & \frac{\partial O}{\partial x_{nn}} \end{pmatrix} \tag{17}$$

Here, $\frac{\partial O}{\partial x_{ij}} = \begin{pmatrix} \frac{\partial o_{11}}{\partial x_{ij}} & \frac{\partial o_{12}}{\partial x_{ij}} & \cdots & \frac{\partial o_{1m}}{\partial x_{ij}} \\ \frac{\partial o_{21}}{\partial x_{ij}} & \frac{\partial o_{22}}{\partial x_{ij}} & \cdots & \frac{\partial o_{2m}}{\partial x_{ij}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial o_{m1}}{\partial x_{ij}} & \frac{\partial o_{m2}}{\partial x_{ij}} & \cdots & \frac{\partial o_{mm}}{\partial x_{ij}} \end{pmatrix}$.

This gives us the expression for $\frac{\partial L}{\partial X}$ as Eq. (18).

$$\frac{\partial L}{\partial X} = \frac{\partial O}{\partial X} * \frac{\partial L}{\partial O}. \tag{18}$$

Here, $*$ represents the convolution operation. Similarly, we know that $\frac{\partial L}{\partial X}$ is solvable. In summary, the proposed QCNN can be updated by using backpropagation.

3.2. Feature fusion

The previously extracted image feature data are fused with the EHR data using direct concatenation, as shown in Fig. 4. Compared with late fusion, early fusion can better utilize the complementary information between various data modalities, and the fused features can be effectively used to train a VQC for intelligent diagnosis. A more formal expression is provided in Eq. (19).

$$y = D\left(F\left(f\left(x_1\right),f\left(x_i\right),\ldots,f\left(x_n\right)\right)\right) \tag{19}$$

where, x_i represents the i-th data source. f represents the data-processing method. F represents the feature fusion method used in this study, and D represents the decision-making method based on the fused features.

3.3. VQC training

Because the previous step in Section 3.2 involved only a simple concatenation operation on the data features, we selected a VQC to learn the relationship between multimodal data and obtain diagnostic results. The training process for a typical VQC is shown in Fig. 5.

First, the fused multimodal data features $\boldsymbol{x}_i$ are encoded into quantum states by using the encoding circuit V to obtain quantum data $|\boldsymbol{x}_i\rangle$, where we employ amplitude encoding. Then, the encoded quantum state is evolved through the quantum circuit $U(\boldsymbol{\theta})$ to obtain the final state $|\psi(\boldsymbol{\theta};\boldsymbol{x}_i)\rangle = U(\boldsymbol{\theta})|\boldsymbol{x}_i\rangle$, where $\boldsymbol{\theta}$ is the parameter set of the entire circuit. Afterward, let us measure a specific qubit of the final state and use the measurement value as the output $\langle\hat{Z}\rangle$, which is also the diagnosis result. Finally, let us calculate the loss $L = loss(\langle\hat{Z}\rangle, y)$ based on the output, and update the parameters $\boldsymbol{\theta}$ according to $\frac{\partial L}{\partial \boldsymbol{\theta}}$ by using classical optimizer. The quantum circuit of the VQC for intelligent diagnosis is shown in Fig. 6, and its specific construction process is shown in Algorithm 2.

Algorithm 2 Construction the quantum circuit for the VQC.

Input: the number of layers l, the input $|\boldsymbol{x}_i\rangle$, the number of qubits n.
Output: Circuit of a variational quantum classifier.

Set $i = 1$;
for $i \leq n$ **do**
  add R_x on qubit i;
  $i = i + 1$
end for
Set $i = 1$;
for $i \leq n$ **do**
  add $CNOT$ gate on qubit i with control qubit $(i+1)\,\%n$
  $i = i + 1$;
end for
Set $i = 1$;
for $i \leq n$ **do**
  add R_y on qubit i;
  $i = i + 1$;
end for
Set $i = 1$;
for $i \leq n$ **do**
  add $CNOT$ gate on qubit $(i+1)\,\%n$ with control qubit i;
end for
Set $i = 1$;
for $i < l$ **do**
  Repeat step 1 to 19;
  $i = i + 1$;
end for
return circuit

4. Experiment

As a general fusion system for intelligent diagnosis, we used the breast cancer diagnosis and Covid-19 diagnosis dataset for experiments to verify the effectiveness of QNMF. To improve the utilization of data and thereby enhance the accuracy of intelligent diagnosis, this study integrated the patient image data and EHR data. We utilized Pennylane to simulate real quantum circuits and extract features from the input images, which were then fused with the EHR data. Subsequently, the VQC was trained using fused features to obtain the diagnosis results. Simultaneously, we conducted two controlled experiments for each experiment: a QCNN with image data and a VQC with EHR data.

4.1. Datasets and initial setting

Owing to the need for privacy protection and scarcity of quantum computing resources, this study simulated the multimodal data of patients using relatively simple publicly available data on the same disease. From a pathological perspective, specific test results are unlikely to exhibit significant differences when different patients have the same condition. Based on this, we can obtain multimodal data that QNMF can effectively process at the current stage. For the breast cancer diagnosis experiment, we selected two datasets: Medminst [47] and Breast Cancer Wisconsin [48]. The first data modality was Medminst's breast dataset, which consisted of breast ultrasound images. The second data modality was an EHR dataset called the Breast Cancer Wisconsin dataset, which comprises features manually extracted from breast cell slices. Both were clinical diagnostic datasets that can be used independently to diagnose breast cancer. However, their combination allowed the creation of multimodal data for breast cancer diagnosis. Similarly, for the diagnosis of Covid-19, we utilized two modalities of data: the Covid-19 CT scan image dataset [49], where all images were sampled into 32×32 pixels, and the Covid-19 blood test features [50], which were also EHR data. We selected 500 labeled data points from the two datasets of each experiment. The QCNNs used in this study are illustrated in Figs. 7 and 8.

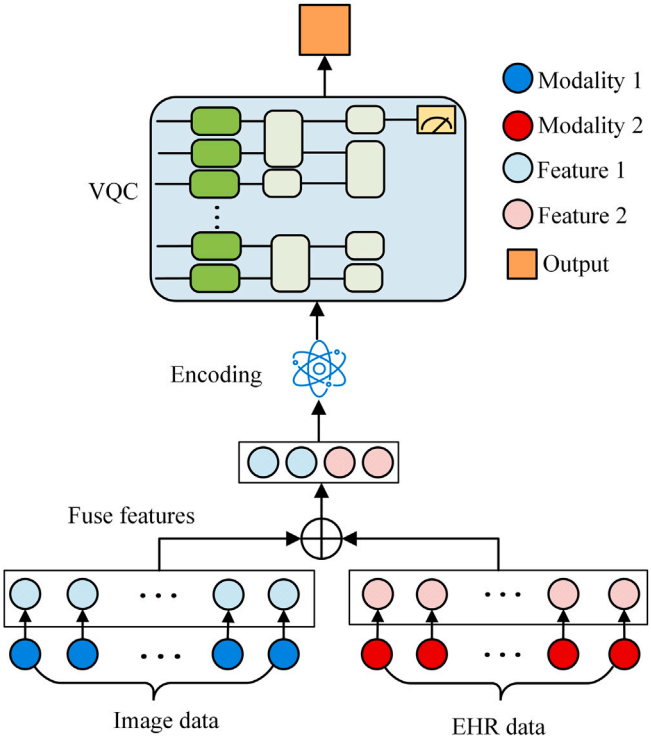


Fig. 4. The early fusion method.

4.2. Evaluation metric

For the controlled experiments, classical ML was used as an evaluation indicator to compare the performances of the experiments and calculate the corresponding accuracy, precision, recall, and F1-score. The metric formulas are given in Eq. (20).

$$\begin{aligned} accuracy &= \frac{TN+TP}{TP+TN+FP+FN} \\ precision &= \frac{TP}{TP+FP} \\ recall &= \frac{TP}{TP+FN} \\ F1-score &= \frac{2*precision*recall}{precision+recall} \end{aligned} \quad (20)$$

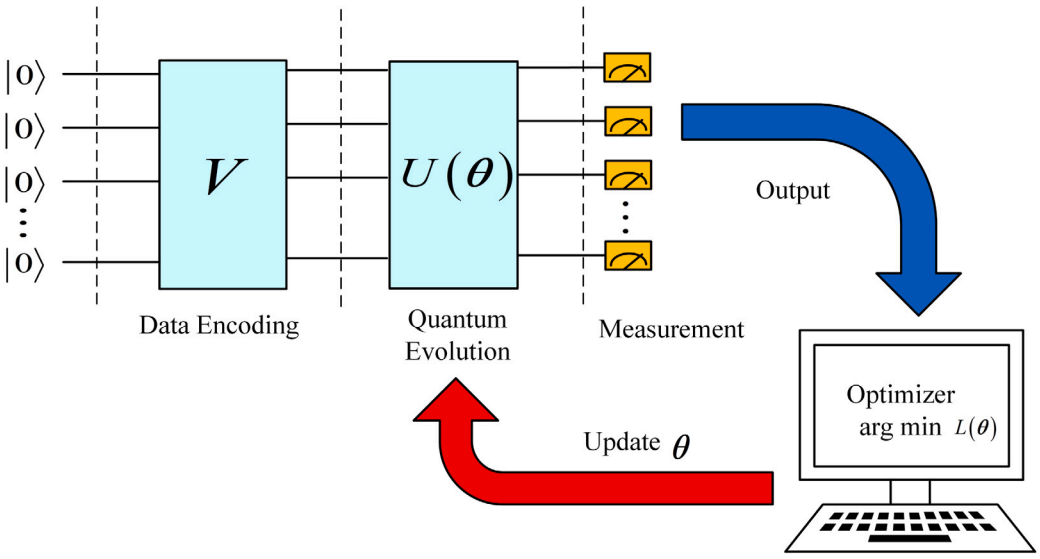


Fig. 5. The training process of a typical VQC.

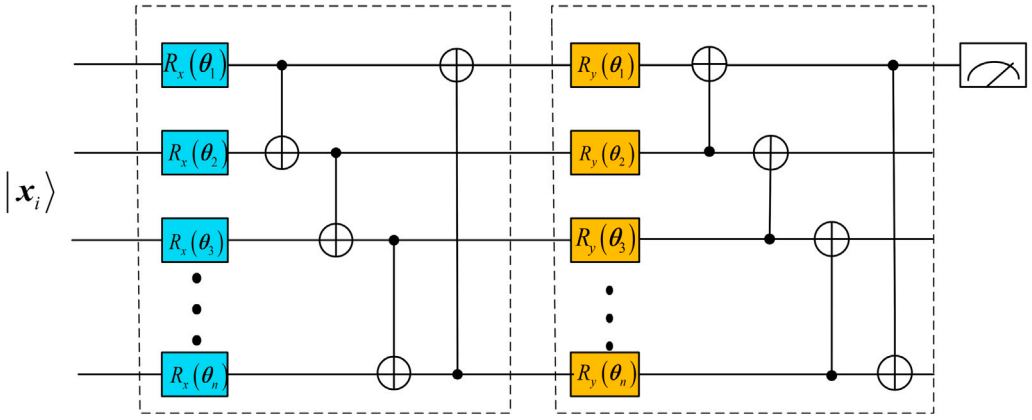


Fig. 6. The quantum circuit used for VQC.

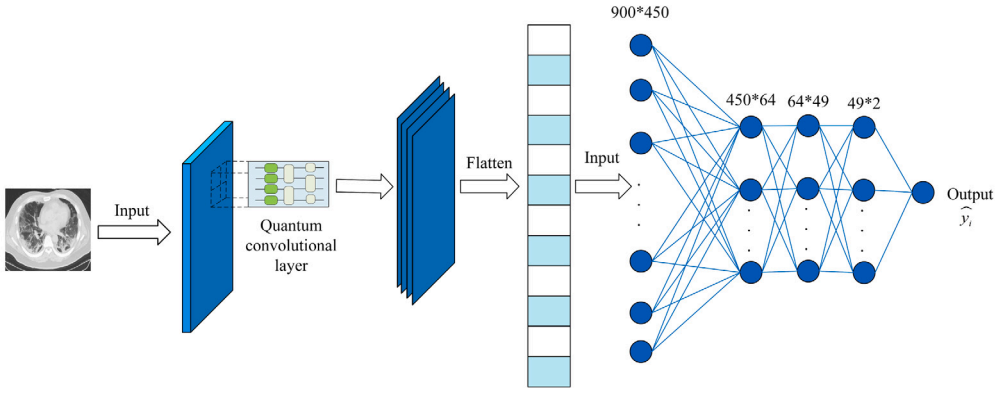


Fig. 7. QCNN for the breast cancer diagnosis.

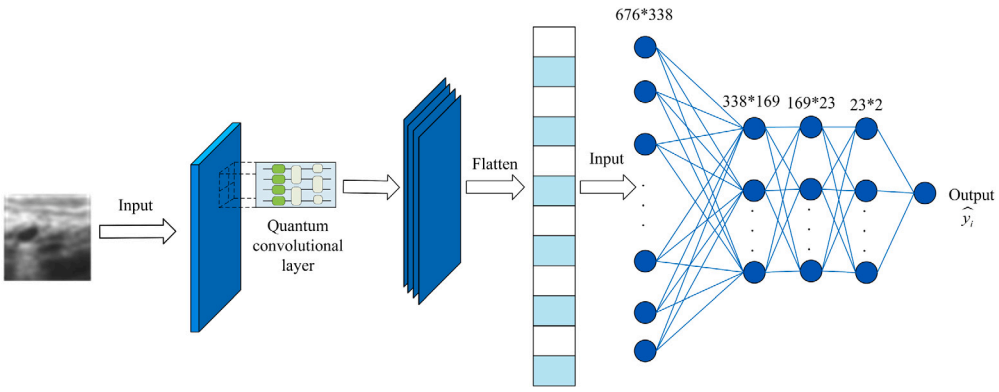


Fig. 8. QCNN for the Covid-19 diagnosis.

Table 1
Hyperparameters of the breast cancer diagnosis experiment.

Learning rate	Batch size	Number of qubits	Epoch
0.1	16	4	100

Table 2
Performance comparison of classical pre-trained neural models and QCNN.

Method	Accuracy	AUC
ResNet-18	**0.8630**	0.9010
Auto-Sklearn	0.8030	0.8360
AutoKeras	0.8310	0.8710
Google AutoML Vision	0.8600	**0.9190**
QCNN	0.8386	0.8049

Table 3
Comparison of related indicators for the three experiments of a breast cancer diagnosis.

Method	Accuracy	Precision	Recall	F1-Score
QCNN with ultrasound images	0.8386	0.4549	0.8971	0.6037
VQC with cell slices features	0.9532	0.9523	0.9478	0.9501
QNMF	**0.9707**	**0.9748**	**0.9747**	**0.9745**

Here, TP represents the situation in which the sample is actually positive and the model also predicts the sample as positive. TN represents a situation in which the sample is actually negative, and the model also predicts the sample as negative. FP represents a situation in which the sample is actually negative, but the model predicts it as positive. FN represents the situation in which the sample is actually positive, but the model predicts it as negative.

4.3. Experiment results and analysis

4.3.1. Breast cancer diagnosis experiment result and analysis

The related hyperparameters of the breast cancer diagnosis experiment are listed in Table 1, and the experimental results are shown in Fig. 9.

First, we compared the classical CNN and QNN to measure the feature extraction performance of QCNN, as shown in Table 2 [47]. This represents the model accuracy, and AUC represents the area under the curve.

Based on the comparison results in Table 2, we conclude that the proposed QCNN exhibits considerable performance in terms of accuracy and AUC for image feature extraction.

As observed from the results in Table 3, the performance of QNMF has higher accuracy, precision, recall, and F1-score than those of the single-modality. In addition, QNMF trained using fused features surpassed existing classical models in terms of accuracy. Similarly, we can draw the following conclusion: Multimodal data fusion can also improve the accuracy of QML models during the NISQ era.

Table 4
Hyperparameters of the Covid-19 diagnosis experiment.

Learning rate	Batch size	Number of qubits	Epoch
0.1	16	4	50

Table 5
Performance comparison of classical pre-trained neural models and QCNN.

Method	Accuracy	AUC
VGG-16	**0.9963**	**0.9961**
ResNet-18	0.9871	0.9877
QCNN	0.9705	0.8786

Table 6
Comparison of related indicators for the three experiments of Covid-19 diagnosis.

Method	Accuracy	Precision	Recall	F1-Score
QCNN with CT scan images	0.9705	0.9384	0.9786	0.9581
VQC with blood test features	0.9338	0.9385	0.8841	0.9104
QNMF	**0.9761**	**0.9538**	**0.9789**	**0.9662**

4.3.2. Covid-19 diagnosis experiment result and analysis

The hyperparameter settings of the Covid-19 diagnosis experiment are shown in Table 4, and the experimental results are listed in Fig. 10.

In addition, we compared the classical CNN and QCNN to measure the feature extraction performance of the QCNN, and the results are listed in Table 5.

Based on the experimental results in Table 5, we can conclude that VGG-16 and ResNet-18 demonstrate high accuracy and AUC values for the current task, indicating that these two classical deep learning models are suitable for image classification tasks. Although the performance of QCNN is slightly inferior in terms of these two indicators, it is based on a quantum computer and has the potential advantages of quantum computing. Quantum computing can quickly and effectively process complex data to extract image features. Therefore, with the further maturity of quantum computing technology and future improvements and optimizations of model structures, the QCNN is expected to achieve better results in image processing tasks. We also calculated the accuracy, precision, recall, and F1-score for the three experiments; the results are provided in Table 6.

Based on the results in Table 6, we conclude that the QCNN using CT scan images can achieve higher accuracy and F1-score. However, the recall rate is relatively low because of the inability to fully utilize other disease-related information. By contrast, VQC with blood test features has a lower accuracy and recall rate, because the blood test features can not provide sufficient information for accurate disease diagnosis. However, QNMF can fully use information from different sources, integrating CT scan images and blood test features, significantly improving the accuracy and F1-score. Therefore, the QNMF is reasonable and effective. Simultaneously, the accuracy gap between

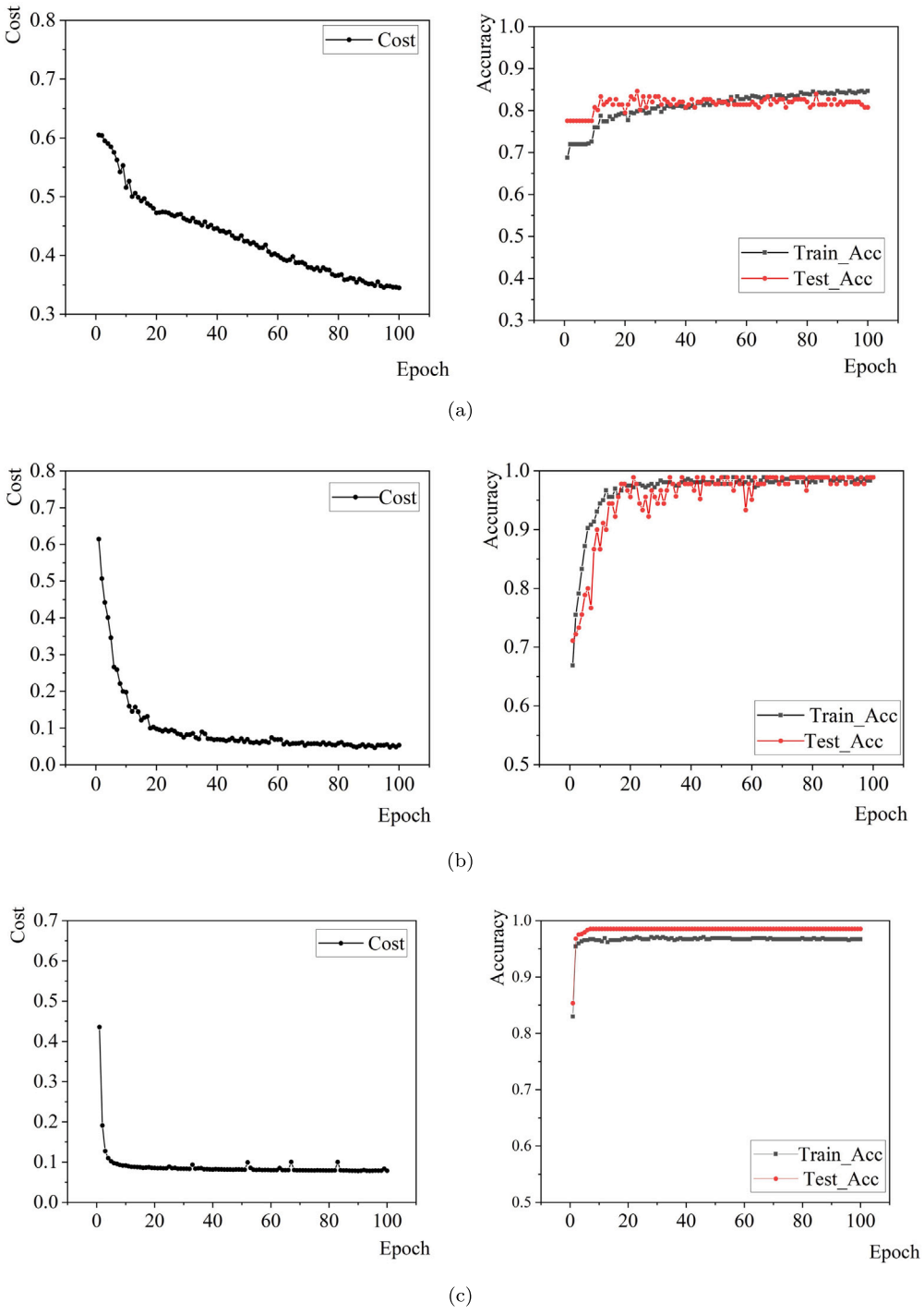


Fig. 9. Cost and Train/Test accuracy for breast cancer diagnosis experiment. (a) Cost and Train/Test accuracy for QCNN with ultrasound images, (b) Cost and Train/Test accuracy for VQC with cell slices features, and (c) Cost and Train/Test accuracy for QNMF.

the QNN and classical models narrowed further when using QNMF, suggesting that data fusion may be a viable strategy for improving QML performance during the NISQ era.

4.4. Resilience analysis to quantum noise

In the NISQ era, any QML model must consider the effects of quantum noise. Quantum noise can severely affect the results of the algorithms, not only reducing the accuracy of the algorithms but also causing the models to fall into a barren plateau and fail to converge. Single-qubit quantum noise can be represented by Kraus matrices, and four typical quantum noises are as follows [3].

Bit flip (BF): The corresponding Kraus matrices are Eq. (21).

$$K_0 = \sqrt{1-p}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad K_1 = \sqrt{p}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{21}$$

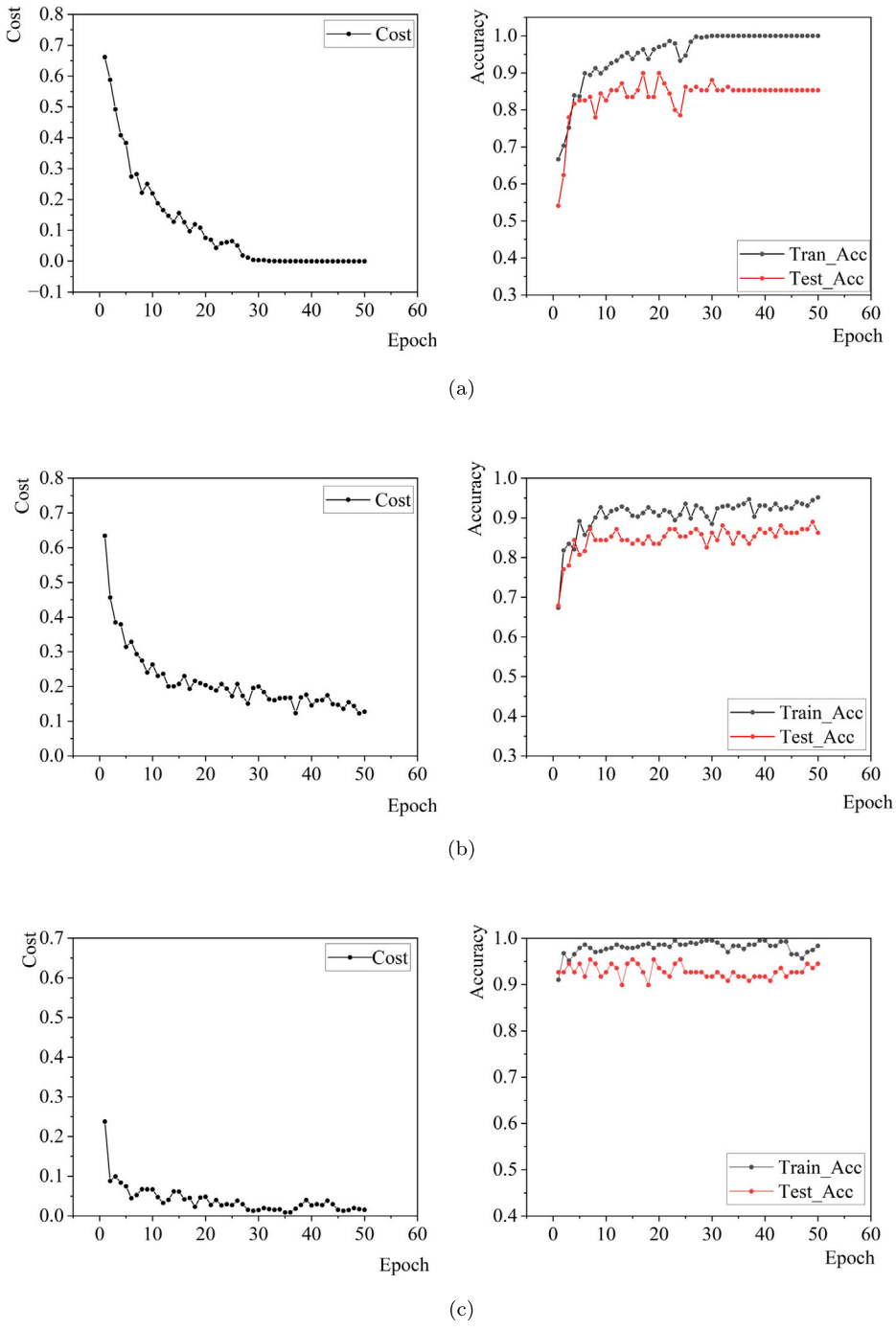


Fig. 10. Cost and Train/Test accuracy for Covid-19 diagnosis experiment. (a) Cost and Train/Test accuracy for QCNN with CT scan images, (b) Cost and Train/Test accuracy for VQC with blood test features, and (c) Cost and Train/Test accuracy for QNMF.

where $p \in (0, 1)$ represents the probability of a qubit undergoing a bit flip.

Phase flip (PF): The corresponding Kraus matrices are Eq. (22).

$$K_0 = \sqrt{1-p}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad K_1 = \sqrt{p}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{22}$$

where $p \in (0, 1)$ represents the probability of a qubit undergoing a phase flip.

Amplitude damping (AD): The corresponding Kraus matrices are Eq. (23).

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix} \tag{23}$$

Table 7
The fidelities of two algorithms under different single-noises when $p = 0.1$. (QCNN: Quantum Convolutional Neural Network, VQC: Variational Quantum Classifier, BF: Bit flip, PF: Phase flip, AD: Amplitude damping, and DN: Depolarizing noise).

Noise type	QCNN	VQC
No noise	1	1
BF	0.9150	0.9965
PF	0.9840	0.9967
AD	0.9818	0.9990
DN	0.9333	0. 9894

Table 8
The fidelities of two algorithms under different single-noises when $p = 0.01$.

Noise type	QCNN	VQC
No noise	1	1
BF	0.9918	0.9999
PF	0.9974	0.9997
AD	0.9932	0.9985
DN	0.9933	0.9980

Table 9
The fidelities of two algorithms under different mixed noises when $p = 0.1$.

Noise type	QCNN	VQC
No noise	1	1
BF and PF	0.8909	0.9915
BF and AD	0.9468	0.9909
BF and DN	0.9217	0.9910
PF and AD	0.8829	0.9985
PF and DN	0.9105	0.9904
AD and DN	0.9285	0.9979

where $\gamma \in (0, 1)$ represents the probability of a qubit undergoing amplitude damping.

Depolarizing noise (DN): The corresponding Kraus matrices are Eq. (24).

$$K_0 = \sqrt{1-p}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$
$$K_1 = \sqrt{p/3}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
$$K_2 = \sqrt{p/3}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad (24)$$
$$K_3 = \sqrt{p/3}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

where $p \in (0, 1)$ represents the probability of a qubit undergoing depolarizing noise.

Fidelity is an essential indicator of the robustness of quantum algorithms to quantum noise. Fidelity measures the closeness of two quantum states and, represents the probability that one state will pass a test to be identified as the other. A good quantum algorithm should have high fidelity, which can also be considered as a measure of the impacts of quantum noise on the algorithm results. The smaller the noise impacts on the results, the higher the fidelity. The fidelity estimation formula for the two mixed states is given by Eq. (25).

$$F(\rho, \sigma) = Tr\left(\sqrt{\sqrt{\rho}\sigma\sqrt{\rho}}\right)^2, \quad (25)$$

where, ρ and σ are the density matrices of the two mixed states. We computed the fidelity of the quantum circuit in the presence of single noise and a combination of two noises for p = 0.1 and p = 0.01. The results are shown in Tables 7, 8, 9, and 10.

According to the results in Tables 7, 8, 9 and 10,under different types of noises, the fidelities of both algorithms decrease but still maintain high values. The results show that the QNMF has good resistance to quantum noise and can maintain high accuracy and robustness in noisy environments. Therefore, the QNMF has good robustness and reliability in practical applications and is capable of coping with different types of quantum noise.

Table 10
The fidelities of two algorithms under different mixed noises when $p = 0.01$.

Noise type	QCNN	VQC
No noise	1	1
BF and PF	0.9857	0.9998
BF and AD	0.9930	0.9996
BF and DN	0.9838	0.9997
PF and AD	0.9920	0.9992
PF and DN	0.9915	0.9998
AD and DN	0.9930	0.9998

In summary, the QNMF proposed in this study demonstrated promising results in breast cancer and Covid-19 diagnosis experiments. The data fusion method further improves the performance of the QNN, making it more competitive with classical neural networks. Moreover, the QNMF is resilient to quantum noise, ensuring its robustness and reliability for practical applications. As quantum computing technology continues to advance, the proposed system is expected to be further optimized and will play a significant role in medical diagnosis and other related fields.

5. Conclusion and future work

This study proposes a quantum neural network-based multimodal fusion system for intelligent diagnosis. QNMF employs quantum convolutional layers to extract features from collected image data and adopts an early fusion method to integrate data with multiple modalities. Subsequently, the fused features are utilized to train the VQC for intelligent diagnosis. The experimental results demonstrate that QCNN can effectively extract image features. QNMF can achieve accuracies of 97.07% and 97.61% for the breast cancer and Covid-19 diagnosis experiments, respectively. Compared with the models using only single-modality data, their classification accuracies and precisions are substantially improved. This study also investigates the robustness of the new algorithm, revealing that QNMF exhibits high fidelity under different typical quantum noises. This indicates that the QNMF possesses strong robustness against quantum noise. However, owing to the limited availability of quantum computing resources, we are currently unable to conduct more complex experiments to further validate the system. Their ability to handle additional data modalities cannot be fully verified. This will be an area of focus for future studies.

So far, the fusion of multimodal data is a crucial step in the future development of intelligent diagnostics. By integrating data from various modalities, QNMF can effectively assist doctors in making accurate diagnoses. In future research, our primary focus will be on the following aspects: First, we would like to construct quantum circuits with greater expressive power to further enhance the performance of QNN. Second, we would like to further investigate the impact of quantum noise and reduce the influence of noise on the QNN.

CRediT authorship contribution statement

Zhiguo Qu: Data curation, Literature analysis, Interpretation of results, Writing – original draft. **Yang Li:** Data curation, Literature analysis, Interpretation of results, Writing – original draft. **Prayag Tiwari:** Data curation, Literature analysis, Interpretation of results, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data is available at https://github.com/WindMoon12/QNMF.

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